FINANCIAL MANAGEMENT

Suggested Answers
Nov-Dec 2022

Answer to the Question# 1(a) (i):

N = 5, I = 12%, PMT = 0.06x Tk 1,000 = Tk 60; FV = Tk 1,000
PV = Tk 783.71
N = 5, I = 12%, PMT = 0.14x Tk 1,000 = Tk 140; FV = Tk 1,000
PV = Tk 1,072.10

Answer to the Question# 1(a) (ii):

Number of Bond: A bonds = TK 20,000/783.71 = 25,520
Number of Bond: B bonds = TK 20,000/1,072.10 = 18,655

Answer to the Question# 1(a) (iii):

Interest income of A = TK 25,520 bonds x Tk 60 = Tk 1,531.20
Interest income of B = TK 18,655 bonds x Tk 140 = Tk 2,611.70

Answer to the Question# 1(a) (iv):

At the end of the 5 years both bonds mature and will sell for par of Tk 1,000
N = 5, I = 10%, PMT = Tk 60, FV = Tk 366.31
Total future cash flows: Tk 366.31 + Tk 1,000 = Tk 1,366.31
N = 5, I = 10%, PMT = Tk 140, FV = Tk 854.71
Total future cash flows: Tk 854.71 + Tk 1,000 = Tk 1,854.71

Answer to the Question# 1(a) (v):

The difference is due to the differences in interest payments received each year. The principal payments at maturity will be the same for both bonds. The yield to maturity of Bond A is 11.76% and the yield to maturity of Bond B is 11.59% with the 10% reinvestment rate for the interest payments. Siam would be better off investing in Bond A. The reasoning behind this result is that for both the bonds the principal is priced to yield 12%. However, Bond B is more dependent upon the reinvestment of the large coupon payment at the yield to maturity to earn the 12% than is the lower coupon payment of Bond A.

Answer to the Question# 1(b):

i. Real GNP was higher than anticipated. Since returns are positively related to the level of GNP, returns should rise based on this factor.
ii. Inflation was exactly the amount anticipated. Since there was no surprise in this announcement, it will not affect Goodwill Drug's returns.
iii. Interest rates are lower than anticipated Since returns are negatively related to interest rates, the lower than expected rate is good news. Returns should rise due to interest rates.
iv. The President's death is bad news, Although the president was expected to retire, his retirement would not be effective for six months. During to period, he would still contribute to Ac firm. His untimely death means that those contributions will not be made. Since he was generally considered an asset to the firm, his death will cause returns to fall. However, since his departure was expected soon, the drop might not be very large.
v. The poor research results are also bad news. Since Goodwill Drug must continue to test the drug, it will not go into production as early as expected. The delay will affect expected future earnings, and thus i will dampen returns now.
vi. The research breakthrough is positive news for Goodwill Drug. Since it was unexpected, it will cause returns to rise.
vii. The competitor's announcement is also unexpected, but it is not a welcome surprise. This announcement will lower the returns on Goodwill Drug.

Answer to the Question# 1(c):

Cash flow is the lifeblood of a business. Every successful company needs a steady source of income and cash on hand to pay bills and make payroll. We found may companies showing profit but experiencing cash follows difficulties indeed.

The following are the four factors that are most often at the root of cash flow problems, as well as advice on how to avoid or overcome them.

1. Not Paying Attention to Expenses

Many businesses end up in a cash flow crunch due to unexpected expenses (for example, costly repairs to equipment, replacing malfunctioning technology or a natural disaster) or too much money going out each month (such as ongoing expenses that have quietly crept up to an unsustainable level).

Resolving a cash flow crisis requires that business leadership take a renewed, vigorous look at their ongoing cost structure.

Every business owner should have a rigorous process in place to track expenses on a monthly basis and project future expenses for the months ahead. A good business accountant can work to make sure the management has an eye on the overall health of the business cash flows and the actions are taken place proactively to face the challenges when arise.

2. Uncertainty about Future Cash Flow

Some businesses are so caught up in the day-to-day grind of getting work done and paying bills that they don't take time to anticipate what's coming in the next few months. Maintaining healthy cash flow requires a long-term vision. Most accounting experts recommend that every business maintain a six-month cash flow projection with expected revenue and expenses, while also adjusting for any seasonal peaks and valleys.

3. Slow-paying Customers

Many cash flow problems are caused by a delay in receivables, such as when a company's customers or clients are slow in paying their bills, owners may not even realize they have a problem.

Business owners need to put consistent policies and procedures in place to ensure that customers pay in a timely fashion. It's especially important to clarify the payment terms and expectations on every invoice, whether that's "payment due within 30 days" or "payment due upon receipt."

4. No Plan for Collections

There comes a point when a slow-paying customer turns into a delinquent customer. What then? Many businesses do not have a standard process in place to collect on unpaid bills. It doesn't have to be complicated or time-consuming; it's often a matter of scheduling collections into the overall sales process and making it part of ongoing daily business operations. If a customer's payments are overdue by a matter of months instead of weeks, it might be time to consider turning over those bills to a collection agency or third-party collections service.

Answer to the Question# 1(d):

To: Finance Director <email id>

From: My email id

Subject: Dividend Policies

Dear Sir,

As required, the following are the types of dividend policies and factors influencing the same.

<u>Dividend policies:</u>

- Residual Dividend Policy: Dividend is to be paid equal to the actual earnings minus the amount of retained earnings necessary to finance the firm’s optimal capital budget.
- Stable Predictable Dividend: Payment of a specific amount as dividend or periodically increase the dividend at a constant rate so that annual dividend is relatively predictable by investors.
- Constant pay-out ratio: Payment of a constant percentage of earnings as dividend each year.
- Low Regular Dividend Plus Extras: A supplemental dividend paid in years when the firm does well and excess funds are available for distribution.

Factors:

- Constraints on dividend payment - covenants if any by the lenders.
- Existing investment requirements of the firm out of retained earnings.
- If alternative sources of capital are not suitable and retained earnings may sometimes be attractive than other means of capital raising.
- Shareholders' desire: current VS future income.
- Perceived riskiness: dividend VS capital gain.
- Tax impact: dividend VS capital gain.
- Impact on socio-economic environment (signaling).
- Obligation of fiscal law on the firm to distribute a minimum amount of dividend.

Regards.

Answer to the Question# 2(a) (i & ii):

REPORT

To: Board of Directors, New Fashion
From: Financial Analyst
Date: Today
Subject: WACC

Introduction
This report provides an overview of New Fashion's cost of funds to be used for investment appraisal purposes;

Present WACC
New Fashion's current WACC is 17.09% calculated as follows:

WACC	Note	MV	% Cost	% Weight	Weighted
Ordinary Shares (ex div)	1	1,500,000,000	21.00%	67.63%	14.20%
Preference Shares (ex div)	2	488,000,000	6.56%	22.00%	1.44%
Debentures at MV	3	230,000,000	13.9%	10.37%	1.44%
Weighted Average Cost of Capita		**2,218,000,000**		**100.00%**	**17.09%**

Note 1) Cost of Equity (Gordon's Growth Model)
[2.5*(1+.1)/(25)] +.1 = 21%

Note 2) Cost of Preference Shares
Interest Payable/Ex Div. Market value = (400 M * 8%) /(122 * 4M) * 100 = 6.56%

Note 3) Cost of Debentures (IRR Calculation)

Year	Value Tk. 000s	Interest Tk. 000s	Tax Relief Tk. 000s	Redeem Tk. 000s	Net C flow Tk. 000s
0	-230,000				-230,000
1		30,000	-7,500		22,500
2		30,000	-7,500	250,000	272,500

	Discount @ 10%			Discount @ 15%		
Year	Net C flow Tk. 000s	D factor	PV Tk. 000s	Net C flow Tk. 000s	D factor	PV Tk.000s
0	-230,000	1	-230,000	-4,500.00	1	-230,000
1	22,500	0.9091	20,455	400.00	0.8696	19,566
2	272,500	0.8264	225,194	5,400.00	0.7561	206,037

Net Present Value ('000) + 15,648.75 **Net Present Value ('000)- 4,396.75**

IRR (Cost of Debentures)=10%+[15,648.75/(15648.75+4396.75)]*(15%-10%) = 13.90%

Proposed WACC

Allowing for an after tax cost of funds of 7.5% (10% - 25% tax relief) New Fashion's WACC would reduce to 16.12% as calculated below. This rate should be used for the purposes of future capital investment appraisal, if no additional funds are raised specifically therefor.

WACC	Note	MV	% Cost	% Weight	Weighted
Ordinary Shares (ex div)		1,500,000,000	21%	60.78%	12.76%
Preference Shares (ex div)		488,000,000	6.56%	19.77%	1.30%
Irredeemable Loan Stock		230,000,000	13.90%	9.32%	1.30%
Debentures at MV		250,000,000	7.50%	10.13%	0.76%
Weighted Average Cost of Capital		**2,468,000,000**	**100.00%**		**16.12%**

Answer to the Question# 2(b):

Briefing Note

To: Board of Directors, Tulip Resort
From: Finance Manager
Date: Today
Subject: Acquisition of Cox's Bazar's Resort – Cash Vs Share Consideration

Advantages of Share Consideration
- Liquidity Maintained
- Tax Efficiency (defer Capital Gain Tax)
- Potential for shareholders to retain stake in the business
- Improves gearing

Disadvantages of Share Consideration
- Control dilution
- Reduces EPS
- Increase future dividend expectation
- Increase future earnings expectation

Advantages of Cash Consideration
- No control dilution
- No reduction in EPS
- Removes potential threat of hostile takeover bid
- More attractive to shareholders of target company

Disadvantages of Cash Consideration
- Liquidity deterioration
- Crystallises Capital Gain
- If raised using debt then increases financial risk

Answer to the Question# 3(a) (i):

Total fund to be invested = (Tk.20m x 5) = Tk.100m.

Justify the investment decision under portfolio theory:

Given that the projects are not divisible and cannot be postponed.

Step I: Expected rate of return (ERR) from the possible investment combinations are as follows.

Project 1 + Project 2 = (0.22 x 0.6) + (0.26 x 0.4) = 0.236 i.e. 23.6%

Project 2 + Project 3 = (0.26 x 0.4) + (0.28 x 0.6) = 0.272 i.e. 27.2%

Project 2 + Project 4 = (0.26 x 0.4) + (0.34 x 0.6) = 0.308 i.e. 30.8%

Project 1 + Money Market (MM) = (0.22 x 0.6) + (0.18 x 0.4) = 0.204 i.e. 20.4%

Project 2 + Money Market (MM) = (0.26 x 0.4) + (0.18 x 0.6) = 0.212 i.e. 21.2%

Project 3 + Money Market (MM) = (0.28 x 0.6) + (0.18 x 0.4) = 0.240 i.e. 24.0%

Project 4 + Money Market (MM) = (0.34 x 0.6) + (0.18 x 0.4) = 0.276 i.e. 27.6%

The investors require a minimum return of 24% (RRR) over the three-year period. However, the following four options are eligible for further analysis to pick the most suitable investment decision in view of minimum RRR = 24%.

Opt-1: Project 2 + Project 3

Opt-2: Project 2 + Project 4

Opt-3: Project 3 + MM

Opt-4: Project 4 + MM

Step II: Determination of Standard Deviations (σ) of above investment combinations.

σ^2 P2+P3 = $\sqrt{(9)^2(0.4)^2 + (15)^2(0.6)^2 + 2\ (0.4)\ (0.6)\ (9)\ (15)\ (0.65)} = 11.67$

σ^2 P2+P4 = $\sqrt{(9)^2(0.4)^2 + (13)^2(0.6)^2 + 2\ (0.4)\ (0.6)\ (9)\ (13)\ (0.57)} = 10.29$

σ^2 P3+MM = $\sqrt{(15)^2(0.6)^2 + (5)^2(0.4)^2 + 2\ (0.6)\ (0.4)\ (15)\ (5)\ (0.55)} = 10.24$

σ^2 P4+MM = $\sqrt{(13)^2(0.6)^2 + (5)^2(0.4)^2 + 2\ (0.6)\ (0.4)\ (13)\ (5)\ (0.60)} = 9.14$

Step III: Coefficient of Variation (CV):

P2+P3 = 11.67 / 27.2 = 0.4290

P2+P4 = 10.29 / 30.8 = 0.3341

P3+MM = 10.24 / 24 = 0.4267

P4+MM = 9.14 /27.6 = 0.3312

In view of the CV calculation, P4+MM is the minimal.

Hence, the investment decision would be Tk. 60 million in Project 4 and Tk. 40 million in money market under Portfolio theory.

Justify the investment decision under CAPM:

Beta (β) of projects: $\sigma_j Cor_{jm} / \sigma_m$

$\beta_{\text{Project 1}} = 7 \times 0.68 / 10 = 0.476$

$\beta_{\text{Project 2}} = 9 \times 0.65 / 10 = 0.585$

$\beta_{\text{Project 3}} = 15 \times 0.75 / 10 = 1.125$

$\beta_{\text{Project 4}} = 13 \times 0.88 / 10 = 1.144$

$\beta_{\text{Project MM}} = 5 \times 0.40 / 10 = 0.200$

Calculation of Required Rate of Return (RRR): $k_{RF} + (k_M - k_{RF})\ \beta_j$

$k_{\text{Project 1}} = 16 + (27 - 16)\ 0.476 = 21.24$

$k_{\text{Project 2}} = 16 + (27 - 16)\ 0.585 = 22.44\%$

$k_{\text{Project 3}} = 16 + (27 - 16)\ 1.125 = 28.38\%$

$k_{\text{Project 4}} = 16 + (27 - 16)\ 1.144 = 28.58\%$

$k_{\text{Project MM}} = 16 + (27 - 16)\ 0.200 = 18.20\%$

Calculation of RRR of possible portfolios

$k_{\text{Project 1+2}} = (21.24 \times 0.6) + (22.44 \times 0.4) = 21.72$

$k_{\text{Project 2+3}} = (22.44 \times 0.4) + (28.38 \times 0.6) = 26.00$

$k_{\text{Project 2+4}} = (22.44 \times 0.4) + (28.58 \times 0.6) = 26.12$

$k_{\text{Project 1+MM}} = (21.24 \times 0.6) + (18.20 \times 0.4) = 20.02$

$k_{Project\ 2+MM}$ = (22.44 x 0.4) + (18.20 x 0.6) = 19.90

$k_{Project\ 3+MM}$ = (28.38 x 0.6) + (18.20 x 0.4) = 24.31

$k_{Project\ 4+MM}$ = (28.58 x 0.6) + (18.20 x 0.4) = 24.43

We will consider the portfolio which has the highest ERR over RRR subject to condition ERR≥24%

$k_{Project\ 1+2}$ = does not fulfill the condition

$k_{Project\ 2+3}$ = 27.2 – 26 = 1.20

$k_{Project\ 2+4}$ = 30.8 – 26.12 = 4.68

$k_{Project\ 1+MM}$ = does not fulfill the condition

$k_{Project\ 2+MM}$ = does not fulfill the condition

$k_{Project\ 3+MM}$ = 24 – 24.31 = (0.31)

$k_{Project\ 4+MM}$ = 27.6 – 24.43 = 3.17

Hence, the investment decision would be Tk. 40 million in Project 2 and Tk. 60 million Project 4 under CAPM.

Answer to the Question# 3(a) (ii):

Under portfolio theory, risk and return of portfolio would be focused instead of analysis of single security. The portfolio variance (standard deviation) is affected by the association of movement of returns of two securities. Investors are risk averse and they hold well diversified portfolios rather investing entire wealth in a single security. On the other hand, CAPM intends to price individual security and determine the required rate of return from individual security assuming the investors prefer the highest expected returns for a given level of risk. For these reasons, the portfolio theory and CAPM might give different solutions in investment decision process.

Answer to the Question# 3(a) (iii):

A portfolio only contains systematic risk when it becomes market portfolio. Market portfolio includes at least one lot of all stocks traded in the market. Thus, market portfolio eliminates all unsystematic risk (avoidable) and it carries only systematic risk (market risk). It is the optimum risk diversification from investor's point of view. However, investor cannot diversify market risk as it is beyond his control so that he demands risk premium thereof.

Answer to the Question# 3(a) (iv):

'Limit Order' means an order to buy or sell stock at a specified price. A limit order sets the maximum price the client is willing to pay as a buyer or the minimum price he is willing to accept as a seller. Thus, it fixes the 'upper' or 'lower' limit of security price accepted by the buyer or seller during trading in the stock market.

Answer to the Question# 3(a) (v):

The bond-yield-plus model is based on the empirical observation that the return on equity is higher than the yield on bonds. Since equities are riskier than bonds, the difference between the two is a reward that the investor requires in order to invest in the riskier asset. Now if this equity market premium was constant, then the required rate of return for equity could simply be calculated by looking at the bond yields and then adding the fixed premium.

Answer to the Question# 3(b):

The price-earnings ratio (P/E) is the market price per share of common stock dividend by the earnings per share. It indicates the amount an investor is willing to pay for each taka of earnings. It is used to assess the owner's appraisal of the value of the firm's earnings. The level of the P/E ratio indicates the degree of confidence that investors have in the firm's future. The market/book (M/B) ratio is the market price per common stock divided by the firm's book value per share. Firms with high M/B ratios are expected to perform better than firms with lower relative M/B values.

Answer to the Question# 3(c):

First need to calculate the forward rate of exchange at which the hedging contract is made.

UK company will receive $500,000 in six months' time.

At the interest rate of 7.5%, this equates to $481,927.71 at today's date ($500,000 / 1.0375).

$481,927.71 at the spot rate of $2.50 equates to Pound 192,771.08

Pound 192,771.08 at an interest rate of 6% would be worth Pound 198,554.21 in six months' time (Pound 192,771.08 x 1.03).

Therefore, the forward rate is 2.5182 (500,000 / 198,554.21).

The effect on UK company in the event of the Pound moving in different ways over the six months' period can now be calculated.

(i) If the Pound gains 4%, the exchange rate will be $2.60 ($2.50 x 1.04). $500,000 would therefore buy only Pound 192,307.69 (500,000 / 2.60). Hedging has saved the company Pound 6,246.52 (Pound 198,554.21 – Pound 192,307.69).

(ii) If the Pound loses 2%, the exchange rate will be $2.45 ($2.50 x 0.98). $500,000 would therefore buy Pound 204,081.63 (500,000 /2.45). Hedging has cost the company Pound 5,527.42 ((Pound 204,081.63 – Pound 198,554.21).

(iii) If the Pound remains at $2.50, the transaction would realize Pound 200,000 (500,000 / 2.50). Hedging has therefore cost the company Pound 1,445.79 (Pound 200,000 – Pound 198,554.21).

Answer to the Question# 4(a):

INDICATIVE CONTRACT PROFIT

Cost of Contract		**Tk.**
10,000 cases * 15 bottles * Tk. 100 per bottle		-80,000
Income	**LP**	
10,000 LP per case * 35,000 cases	350,000,000	
Bank Buy Rate	17	
Deduct: 6 Month Premium	1	
LP Six Month Forward Rate	**16**	
Taka Equivalent		21,875,000
Indicative Profit on Transaction		**6,875,000**

Answer to the Question# 4(b):

- Credit vetting of customers (rating agencies)
- Regular review of accounts of customers
- Government Export Guarantees
- Export Credit Insurance

Answer to the Question# 5(a) (i):

Valuation of N Ltd:

Market capitalization of N Ltd:

Value of ordinary shares in statement of financial position = Tk.20·0 million
Nominal (par) value of ordinary shares = Tk.0.5
Number of ordinary shares of company = 20m/0·5 = 40 million shares
Ordinary share market price = Tk.4·00 per share
Market capitalization = 40m x 4 = Tk.160 million

Answer to the Question# 5(a) (ii):

Valuation of N Ltd:

Net asset value (liquidation basis):

Current net asset value (NAV) = 91·0m + 8·3m – 7·1m – 25·0m = Tk.67·2 million
Decrease in value of non-current assets on liquidation = 86·0m – 91·0m = Tk.5 million
Increase in value of inventory on liquidation = 4·2m – 3·8m = Tk.0·4 million
Decrease in value of trade receivables = 4·5m x 0·2 = Tk.0·9 million
NAV (liquidation basis) = 67·2m – 5m + 0·4m – 0·9m = Tk.61·7 million

Answer to the Question# 5(a) (iii):

Valuation of N Ltd:

Price/earnings ratio value:

Historic earnings of N Ltd = Tk.10·1 million
Average price/earnings ratio of N Ltd business sector = 17 times
Price/earnings ratio value of N Ltd = 17 x 10·1m = Tk.171·7 million

Answer to the Question# 5(a) (iv):

Valuation of N Ltd:

Dividend growth model value:

- Using historic dividend growth rate:

  Historic dividend growth rate = [(6·0m/5·0m)1/3 – 1] x 100 = 6·27%

  An assumption is made that future dividend growth is similar to historic dividend growth.

  Value of N Ltd = (6m x 1·0627)/(0·09 – 0·0627) = Tk.234 million

- Using Gordon's growth model:

  Gordon's growth model estimates the dividend growth rate using g = br

  Retention ratio (b) = [(10.1 – 6.0) / 10.1] x 100 = 40.59%

  Current return on shareholders' funds (r) = (10.1 / 67.2) x 100 = 15.03%

  Dividend growth rate (br) = 0.4059 x 0.1503 = 0.061

  Value of N Ltd = (6m x 1·061)/(0·09 – 0·061) = Tk.219.52 million.

Answer to the Question# 5(b):

Theoretical ex-rights value per share:

[(8,000,000 X 15) + (2,000,000 X 12)] / (8,000,000 + 2,000,000) = Tk.14.40.

Answer to the Question# 5(c):

The main drivers of option value are:

1. Value of the underlying: For quoted underlying a mid price of the day is usually used for option pricing. In the case of unquoted underlying assets a separate exercise must be undertaken to value them e.g. using P/E ratio to value the share.
2. Exercise price: It must be mentioned in the option contract. Exercise price is the projected future value of underlying assets agreed by the option writer and option holder.
3. Time to expiry: Longer expiration time reveals higher risk due to volatility in price as a consequence higher premium and vice versa. Hence, option value largely depends on the size of expiration period.
4. Volatility: Option price is also influenced by the volatility of price of underlying asset. If the degree of volatility remains higher, the option price would be higher and vice versa.
5. Risk free rate: This is the minimum return required by the investors from a risk-free investment. It covers inflation and maintains purchasing power of money over the periods.

Answer to the Question# 6(a):

Year	Items	Amount	PV Factor @12%	PV
0	Purchase of Machinery	(705,000,000)	1	(705,000,000)
01-04	Annual contribution margin (1,500,000-1,200,000)*1200	360,000,000	3.037	1,093,320,000
01-04	Incremental Fixed cost (146.25 m-11.25 m-37.5 m)	(97,500,000)	3.037	(296,107,500)
4	Sales of Machinery	120,000,000	0.636	76,320,000
NPV of producing vehicle				168,532,500
NPV of selling the patent				150,000,000
Excess NPV of producing the product				**18,532,500**

Answer to the Question# 6(b) (i, ii, iii & iv):

Sensitivity analysis

Initial outlay:

$= \frac{168,532,500}{705,000,000}$ X 100 = 23.91%

Discount rate:

Have to find IRR of the project at first.

Year	Items	Amount	PV Factor @12%	PV	Amount	PV Factor @25%	PV
0	Purchase of Machinery	(705,000,000)	1	(705,000,000)	(705,000,000)	1	(705,000,000)
01-04	Annual contribution margin (1,500,000-1,200,000)*1200	360,000,000	3.037	1,093,320,000	360,000,000	2.362	850,320,000
01-04	Incremental Fixed cost (146.25 m-11.25 m-37.5 m)	(97,500,000)	3.037	(296,107,500)	(97,500,000)	2.362	(230,295,000)
4	Sales of Machinery	120,000,000	0.636	76,320,000	120,000,000	0.41	49,200,000
	NPV			**168,532,500**			**(35,775,000)**

By using extrapolation, IRR can be found as:

$12\% + \frac{168,532,500}{168,532,500+35,775,000}$ (25% - 12%)

= 12% + 10.72%

= 22.72% ≈ 22.7% (appx.)

Therefore sensitivity is ((22.72/12)-1) = 89.3%

Residual value:

$= \frac{168,532,500}{76,300,000}$ X 100 = 220.82%

Annual net operating cash flow:

Sensitivity = $\frac{168,532,500}{(1,093,320,000-296,107,500)}$ X 100 = 21.14%

Answer to the Question# 6(c):

Evaluation

Finding in (a)

After comparing the NPV of selling the product and selling the patent, it is found that selling the patent immediately has Tk. 18,532,500 less NPV than the option of producing. Thus, based on this financial information only, the product should be produced.

Finding in (b)

Initial outlay must increase by 24% to make the NPV zero (moderate risky).

Discount rate need to be increased by 89% to make the NPV zero (low risk).

Residual value need to be decreased by 220.82% to make NPV zero (low risk).

Annual net operating cash flow need to decrease by 21% to make NPV zero (moderate risk).

Therefore, based on the sensitivity overall risk of the estimated NPV is said to be low

---The End---